

LINEAR DISCRIMINANT ANALYSIS

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Agenda

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- Recap of PCA
 - ▣ Major Limitations
- Dimensionality Reduction in Classification
 - ▣ Key Requirements
- Let's see how PCA fails!
- Linear Discriminant Analysis (LDA)
- PCA versus LDA

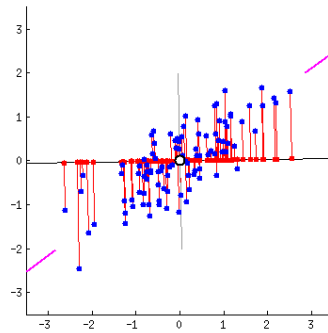
LDA

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PCA: A Quick Recap

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- A dimensionality reduction technique
 - ▣ Finds the most **accurate lower dimensional** representation based on **maximum variance directions**



LDA

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PCA: A Quick Recap

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- Key Features
 - ▣ **Unsupervised** (does not utilize labels)
 - ▣ Focuses on **variance** (not always useful or relevant)

LDA

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PCA: Limitations

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- Limitations
- Not suitable for discrimination/classification
 - ▣ Does not take **class information** into consideration
 - ▣ PCA is an **optimal** dimensionality-reduction technique for **data representation**, not for discrimination/classification problems

LDA

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Dimensionality Reduction in Classification



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- A simple real-world classification example: Identifying species of a fish on a conveyor belt
 - ▣ Species: **Sea bass and salmon**

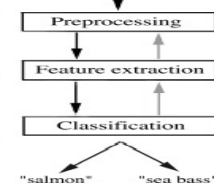


Image source: Pattern Classification by Duda, Hart and Stork

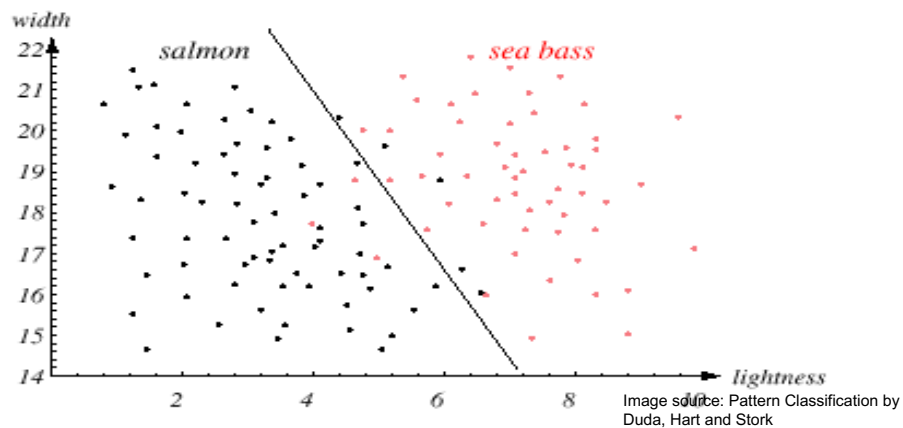
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Dimensionality Reduction in Classification



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- Two features for classification



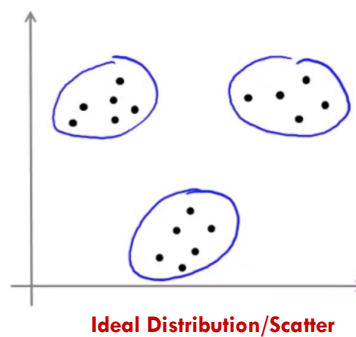
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Dimensionality Reduction in Classification



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- Key Requirements for Classification
 - Each class is as **tight** (compact) as possible
 - Their centroids are as **far** from each other as possible



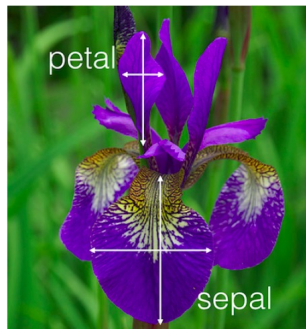
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Let's see how PCA Fails!



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- PCA is not an effective **dimensionality technique for classification**
- The **Iris Dataset**



	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

One of the earliest known datasets used for evaluating classification methods, widely used in ML

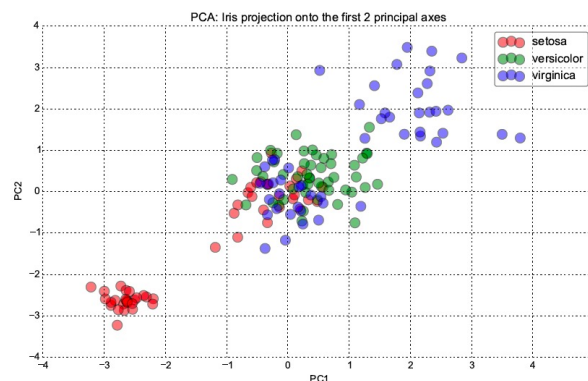
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Let's see how PCA Fails!



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- **Problem:** Given a new measurement of these features, predict the Iris species based on a projection onto a low-dimensional space



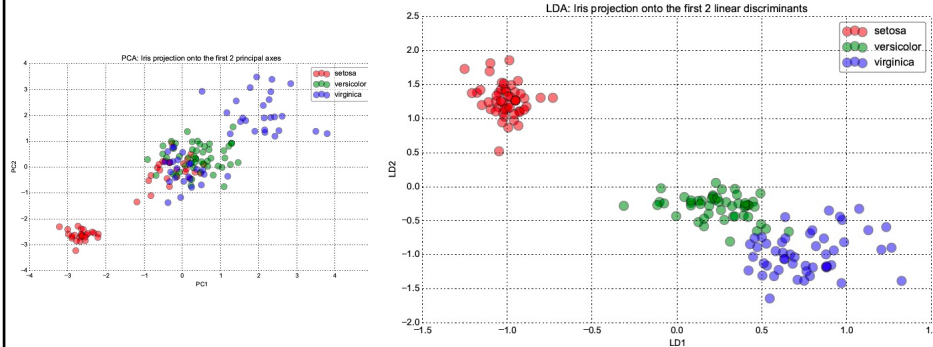
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Let's see how PCA Fails!



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- **Conclusion:** PCA may not be ideal to separate the classes well

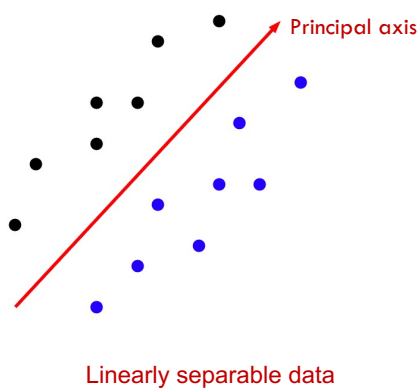


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Why PCA Fails



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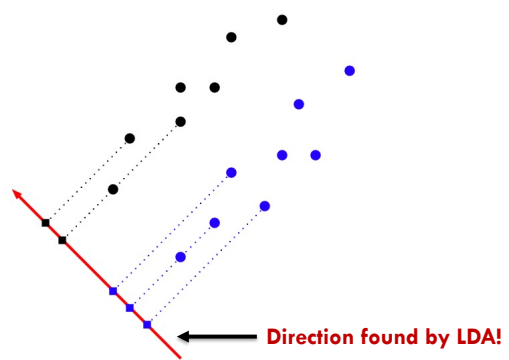
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Linear Discriminant Analysis (LDA)



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- Solution: A DR technique that tries to **preserve the discriminatory information** between different classes of the data set



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Linear Discriminant Analysis



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- Key Idea
 - Select the **best direction for projection** that
 - **Maximizes** their **between-class separability** while
 - **Minimizing** their **within-class variability**
- Key Requirements for Classification

- Each class is as **tight** (compact) as possible
 - Their centroids are as **far** from each other as possible

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